Yizheng Xie

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Education

Ph.D. in Computer Science, Distributed Systems, Operating Systems, Cloud Computing — Sept 2023 – Present
Brown University, Providence, RI, USA (Advisor: Nikos Vasilakis)

M.S. in Computer Science, Database Systems, Streaming Systems — Sept 2021 – May 2023
Boston University, Boston, MA, USA (Advisors: Vasiliki Kalavri, John Liagouris)

B.E. in Electronic Information Engineering, Computer Systems, Machine Learning — Sept 2017 – May 2021
Chinese University of Hong Kong, Shenzhen, Shenzhen, Guangdong, China

Working Experience

Meta Platforms, Menlo Park, CA — May 2022 – Aug 2022
Production Engineer Intern in Kernel Team

- Improved and integrated Fedora and CentOS Stream forges, issue trackers, and build workflow into Meta internal tools.
- Implemented a CLI tool to automate the recursive updating of Rust RPM packages, optimizing parallel chain-build order and ensuring full compatibility (Pagure Repository: pagure.io/fedora-rust/rust-update-set).

Publications

[1] **Yizheng Xie**, Di Jin, Oğuzhan Çölkesen, Vasiliki Kalavri, John Liagouris, and Nikos Vasilakis. *Slowpoke: End-to-end Throughput Optimization Modeling for Microservice Applications*. In 23rd USENIX Symposium on Networked Systems Design and Implementation **(NSDI)**, Renton, WA, May 2026. USENIX Association. (Paper, Code)

[2] Zhicheng Huang, Ramiz Dundar, **Yizheng Xie**, Konstantinos Kallas, and Nikos Vasilakis. *Fractal: Fault-Tolerant Shell-Script Distribution*. In 23rd USENIX Symposium on Networked Systems Design and Implementation **(NSDI)**, Renton, WA, May 2026. USENIX Association. (Paper, Code)

[3] **Yizheng Xie**, Di Jin, and Nikos Vasilakis. *Towards Hybrid Cooperative-Preemptive Scheduling*. In Proceedings of the 13th Workshop on Programming Languages and Operating Systems **(PLOS)**, Seoul, Republic of Korea, October 2025. ACM. (Paper)

Research

Slowpoke: End-to-end Throughput Optimization Modeling for Microservices — Sept 2023 – Apr 2025
To appear at NSDI'26

- Proposed a novel linear programming-based performance model to quantify end-to-end throughput improvements of hypothetical microservice optimizations without requiring white-box knowledge.
- Developed the first system to accurately predict throughput in complex microservices, achieving RMSE of 2.1% across several real-world benchmarks and 1.9% across 108 synthetic benchmarks with diverse characteristics.

Fractal: Fault-Tolerant Shell-Script Distribution — Sept 2023 – Dec 2024
To appear at NSDI'26

- Developed a system for fault-tolerant distribution of unmodified shell scripts, delivering average speedups of 9.6× over Bash, 5.5× over Hadoop Streaming (AHS), and 1.2× over DiSh, the state-of-the-art shell scaleout system.
- Designed a lightweight fault-recovery mechanism—including remote-pipe instrumentation and dynamic result persistence—that achieves recovery within 1.26× of fault-free execution and delivers a 9.3× speedup over AHS.

λeash: Scaling Out Shell Scripts with Resumable Cloud Computing — Sept 2023 – Jan 2025
Under submission (ACM EuroSys'26), with Nikos Pagonas, Haoran Zhang and Konstantinos Kallas

- Developed a system to scale out unmodified shell scripts using cloud (serverful and serverless) computing, achieving 12.7× average and 154.2× peak speedups over Bash; in some cases both faster and cheaper, *e.g.*, achieving a 6.5× speedup while reducing cost by 16.4% compared to PaSh—the state-of-the-art shell parallelization system.
- Designed a *suspend-and-resume* protocol that extends AWS Lambda beyond the 15-minute execution limit while preserving correctness and Unix-style streaming efficiency, with negligible slowdown (avg: 3%).

Incr: Faster Incremental Execution, Opaquely — Aug 2025 – Present
In preparation (OSDI'26)

- Developing a runtime system that automatically memoizes transient results and traces system calls to capture fine-grained and side-effectful dependencies, enabling orders-of-magnitude faster incremental development while providing correctness guarantees.

MeshBench: Benchmarking and Analyzing Service Meshes for Microservices Aug 2025 – Present
In preparation (OSDI'26), with Alice Song, Deepti Raghavan and Akshay Narayan

- Developing a comprehensive benchmarking suite to evaluate the performance overhead and trade-offs of leading service meshes—*e.g.*, Istio, Cilium, Ambient, Canal—in microservice environments, and analyzing how service mesh architectures and functionalities, *e.g.*, security and observability, affect performance and resource utilization.

Invited Talk

Towards Automated Performance Optimizations for Service-Oriented Architectures Oct 2025
Boston University, Boston, MA, USA

Teaching

CSCI 1380 Distributed Systems, Brown University, *Teaching Assistant (Spring 2023, Spring 2024)*

- Co-designed and implemented hands-on assignments for building a large-scale search engine that emphasized core distributed systems concepts, including a distributed key-value store for scalable data storage, data sharding for load balancing, consensus protocols for data consistency, and a MapReduce system for efficient data processing.

CSCI 2952R Systems Transforming Systems Seminar, Brown University, *Teaching Assistant (Fall 2024)*

- Mentored students in conducting state-of-the-art systems research across operating systems, distributed systems, and cloud computing, guiding them through system implementation, experimental evaluation, and academic paper writing.

Skills

- **Programming Languages:** Go, Python, C/C++, Rust, Java, JavaScript, Shell, SQL
- **Systems:** Kubernetes, Docker, Istio, AWS (*e.g.*, Lambda), Linux/Unix, Apache Flink, Spark, V8/libUV, MongoDB, Redis
- **Tools:** Git, CI/CD (GitHub Actions, GitLab CI), Helm, Make, CMake, LaTeX, Vim, perf

Languages and Interests

- **Languages:** Mandarin (native), English (fluent)
- **Interests:** Photography, Video Editing, Badminton, Basketball, Traveling, East Asian History